

Jointly estimating transmissibility and prior immunity from epidemic time series

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Abstract

Infectious disease time series are often used to estimate a pathogen’s basic reproduction number, $\mathcal{R}_0$. However, fits of epidemic models to time series conflate pathogen transmissibility with pre-existing population immunity, so only the *effective* reproduction number, $\mathcal{R}_{\text{eff}}$, can be inferred. This composite parameter is the product of the underlying $\mathcal{R}_0$ and the pre-epidemic susceptible fraction, x^- . We show that a conservation law associated with *epidemic momentum*—prevalence weighted by potential to infect—makes it possible to disentangle transmissibility from prior immunity and to infer $\mathcal{R}_0$ and x^- separately from a single epidemic time series. We test the methodology using stochastic epidemic simulations, and illustrate the approach with a reappraisal of influenza transmissibility during the 1918 pandemic, estimating rather than assuming the degree of prior population immunity. For the autumn wave in Philadelphia, USA, we find $\mathcal{R}_0 \approx 2.7$ and $x^- \approx 0.8$, implying that about 20% of the population was already immune before that wave, plausibly as a result of infection during the spring 1918 herald wave.

Keywords: epidemic momentum, prior immunity, epidemic inference, basic reproduction number, 1918 influenza pandemic

1 Introduction

Epidemic time series are commonly used to infer how well an infection spreads, as measured by the basic reproduction number, $\mathcal{R}_0$ —the expected number of secondary infections caused by a typical infected individual in a fully susceptible population. A standard approach estimates the initial exponential growth rate and combines it with an estimated or assumed distribution of intrinsic generation intervals—the delays between infection of an individual and the secondary infections caused by that individual—to obtain a reproduction number [1, 2, 3]. More precisely, if x^- is the fraction of the population that was susceptible before the epidemic began, initial growth identifies the effective reproduction number $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$,

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rather than $\mathcal{R}_0$ [4]. Early growth alone does not separate pathogen transmissibility from pre-existing population immunity, because different combinations of $\mathcal{R}_0$ and x^- can give the same $\mathcal{R}_{\text{eff}}$.

The fraction of the population that was already immune when the epidemic began is $z^- = 1 - x^-$. Allowing for pre-existing immunity ($x^- < 1$) changes both the transmissibility inferred for the pathogen and the estimated proportion infected during the focal epidemic, $z^+ = x^- - x^+$, where x^+ is the fraction susceptible after the epidemic. Thus, assuming that the population was initially fully susceptible confounds the interpretation of transmissibility, prior immunity, and final size.

Epidemic momentum [5] provides the additional information needed to separate $\mathcal{R}_0$ from x^- . Conceptually, epidemic momentum measures the infectious potential carried by the population, weighting past infections by their remaining potential to transmit. It can be constructed from an incidence time series and the intrinsic generation-interval distribution $g(\alpha)$, where α is the time since infection. The susceptible fraction and epidemic momentum satisfy a conservation law [5, Section 3(d)]; the value of the conserved quantity is $\hat{y}$, the maximum epidemic momentum. Together with $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$, an estimate of $\hat{y}$ supplies the additional relation needed to determine $\mathcal{R}_0$ and x^- separately.

In this paper, we exploit epidemic momentum to infer $\mathcal{R}_0$ and x^- separately from the same epidemic time series. We assess this momentum-based method for inferring transmissibility and prior immunity using stochastic epidemic simulations for which both quantities are known, and then apply it to the autumn wave of the 1918 influenza pandemic in Philadelphia. The application illustrates the biological consequences of estimating, rather than assuming, the population immunity present before an epidemic.

The inference proceeds from an observed incidence time series, or from a time series from which incidence can be reconstructed (*e.g.*, mortality [6]), together with an assumed or estimated intrinsic generation interval distribution $g(\alpha)$. Initial exponential growth gives $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$. The incidence history and $g(\alpha)$ then determine the epidemic momentum through time, from which we extract its maximum $\hat{y}$. Combining $\hat{y}$ with the early growth estimate separates $\mathcal{R}_0$ from x^- ; prior immunity and final size follow from the inferred susceptible fractions.

We first introduce the renewal-equation formulation of epidemic models, from which we derive the relations underlying the momentum-based method. We then assess its accuracy using stochastic simulations, apply it to main-wave data recorded in Philadelphia in 1918, and discuss the biological interpretation and limitations of the inferences.

2 Methods

2.1 Kermack and McKendrick’s renewal equation

We formulate the momentum-based method using the renewal equation, which represents a broad class of epidemic models without requiring a particular compartmental structure [7, 8, 9]. This formulation is particularly convenient for inference because it is written in terms of incidence—the quantity observed, or reconstructed, from an epidemic time series—and the intrinsic generation-interval distribution, which must already be estimated or assumed

to convert an epidemic growth rate into a reproduction number [1].

We measure time in units of a convenient epidemiological timescale, often the mean infectious period, so that τ is dimensionless and the equations can be written in terms of $\mathcal{R}_0$ rather than separate transmission and recovery-rate parameters. We write $X(\tau)$ for the susceptible fraction at time τ , and $\iota(\tau)$ for the incidence, the fraction of the population newly infected per unit time. The probability density $g(\alpha)$ specifies the distribution of the ***intrinsic generation interval***, the delay between infection of an individual and a secondary infection caused by that individual [10, 11]. In dimensionless time, the ***renewal equation*** is

$$\frac{dX}{d\tau} = -\iota(\tau), \quad (2.1a)$$

$$\iota(\tau) = \mathcal{R}_0 X(\tau) \int_{-\infty}^{\tau} \iota(s) g(\tau - s) ds. \quad (2.1b)$$

The first equation identifies incidence with the rate at which susceptibles are depleted. In the second equation, infections that occurred at time s contribute to incidence at time τ according to the generation-interval density evaluated at their infection age, $\tau - s$. The factor $\mathcal{R}_0 X(\tau)$ is the effective reproduction number at time τ .

In the framework represented by Eq. (2.1), differences in infection-age structure among epidemic models are captured by $g(\alpha)$. Standard examples of generation-interval distributions, including those corresponding to the SIR and SEIR models, are given in Table 1. Other choices of $g(\alpha)$ represent models with more general infection-age structure [9]. A fuller account of the renewal equation, its initial history, and its relation to compartmental models is given in [5, Section 2(e)].

The lower limit of integration in Eq. (2.1b) makes explicit that incidence at time τ depends on the preceding incidence history. This history enters the inference in two ways. First, exponentially growing incidence in the early epidemic tail yields the Euler–Lotka relation that identifies $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$. Second, the incidence history and $g(\alpha)$ determine epidemic momentum, which weights past infections by their remaining reproductive potential. In practice, observations begin at a finite time, but the early exponential approximation also allows the unobserved part of the incidence history to be included in the momentum calculation. We begin with the relationship between the asymptotic exponential growth and decay rates and transmissibility.

2.2 Tail exponents and reproduction numbers

In the initial ($\tau \rightarrow -\infty$) and final ($\tau \rightarrow +\infty$) phases of an epidemic, incidence grows and decays exponentially,

$$\iota(\tau) \propto e^{\lambda^\pm \tau}, \quad (2.2)$$

where $\lambda^- > 0$ applies near the start of the epidemic and $\lambda^+ < 0$ applies near the end; we refer to $\lambda^\pm$ as the ***tail exponents***. During these early and late phases, incidence is small and the susceptible fraction is nearly constant, at x^- and x^+ , respectively. Substituting Eq. (2.2) into Eq. (2.1b), rearranging, and cancelling common factors gives the Euler–Lotka equation [12, 13, 14],

$$\frac{1}{\mathcal{R}_0 x^\pm} = \int_0^\infty e^{-\lambda^\pm \alpha} g(\alpha) d\alpha \equiv \mathcal{L}[g](\lambda^\pm) \equiv \mathcal{L}_\pm. \quad (2.3)$$

For the initial growth phase, setting $x^- = 1$ in Eq. (2.3) gives the formula of Wallinga and Lipsitch [1, Equation (2.7)], which is often used with empirical estimates of λ^- to infer $\mathcal{R}_0$ [1, 2, 3]. If $x^- < 1$, the same calculation instead yields $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$ [4].

Estimating x^- empirically is sometimes possible (*e.g.*, [15]). Computationally demanding and/or model-specific methods that attempt to infer or constrain x^- indirectly from the observed epidemic data have also been proposed (see Section 4.1). Our approach obtains the additional information directly from epidemic momentum, which we now define precisely.

2.3 Constructing epidemic momentum from incidence

At infection age α , the upper tail of the generation-interval distribution is the fraction of an individual's intrinsic transmission potential that remains. Multiplying this fraction by $\mathcal{R}_0$, we define the **reduced reproduction number**

$$\mathcal{R}_\alpha = \mathcal{R}_0 \int_\alpha^\infty g(\alpha') d\alpha'. \quad (2.4)$$

Thus $\mathcal{R}_\alpha$ is the expected number of future secondary infections caused by an individual of infection age α in a fully susceptible population, while $\mathcal{R}_\alpha/\mathcal{R}_0$ is the fraction of that individual's reproductive potential that remains (which depends only on $g(\alpha)$, not $\mathcal{R}_0$).

At time τ , the cohort infected at time $\tau - \alpha$ contributes $\iota(\tau - \alpha) d\alpha$ to the infection-age distribution. Weighting each cohort by its remaining reproductive potential we obtain the **epidemic momentum** [5],

$$Y(\tau) = \int_0^\infty \iota(\tau - \alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (2.5)$$

For the SIR and SEIR models, epidemic momentum coincides with the total prevalence of infection. More generally, epidemic momentum is a weighted prevalence: each currently infected individual contributes according to their remaining potential to infect, so individuals at different infection ages carry different weights.

An equivalent representation is useful when epidemic observations are expressed as cumulative incidence. Writing $\bar{\iota}(\tau) = \int_{-\infty}^\tau \iota(s) ds$ and integrating Eq. (2.5) by parts, using $\frac{d}{d\alpha}(\mathcal{R}_\alpha/\mathcal{R}_0) = -g(\alpha)$, gives

$$Y(\tau) = \bar{\iota}(\tau) - \int_0^\infty \bar{\iota}(\tau - \alpha) g(\alpha) d\alpha. \quad (2.6)$$

Thus epidemic momentum can be computed either directly from incidence using Eq. (2.5), or from the corresponding cumulative-incidence curve using Eq. (2.6).

Both representations formally involve the complete incidence history. In practice, suppose observations begin at time τ_i during the initial exponential-growth phase, with $\iota(\tau_i) = \iota_i$. Incidence before τ_i can then be approximated by $\iota(s) \approx \iota_i e^{\lambda^-(s-\tau_i)}$. Splitting Eq. (2.5) into the contributions before and after τ_i gives, for $\tau \geq \tau_i$,

$$Y(\tau) \approx \iota_i e^{\lambda^-(\tau-\tau_i)} \int_{\tau-\tau_i}^\infty e^{-\lambda^-\alpha} \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha + \int_0^{\tau-\tau_i} \iota(\tau - \alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (2.7)$$

The first term in Eq. (2.7) reconstructs the contribution from infections that occurred before observations began, while the second is computed directly from the observed incidence history. A Laplace-transform representation of the first term, including the explicit SEIR expressions used in our calculations, is derived in Section A.

Consequently, the observed incidence curve, its initial growth rate λ^- , and the generation-interval distribution $g(\alpha)$ determine the epidemic momentum throughout the observed epidemic and, in particular, its maximum $\hat{y}$. We next combine this information with $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$ to infer $\mathcal{R}_0$ and x^- separately.

2.4 Inferring $\mathcal{R}_0$ and prior population immunity using epidemic momentum

The preceding subsections provide the two quantities needed to separate $\mathcal{R}_0$ from x^- . First, the rising tail exponent λ^- and the generation-interval distribution $g(\alpha)$ determine $\mathcal{L}_-$ and hence $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^- = 1/\mathcal{L}_-$ [Eq. (2.3)]. Second, the incidence history and $g(\alpha)$ determine the epidemic momentum $Y(\tau)$ [Eq. (2.5)] and hence its maximum $\hat{y}$. We now use the conservation law presented in Ref. [5] to infer $\mathcal{R}_0$ from $\hat{y}$ and $\mathcal{L}_-$, and then infer the pre-epidemic susceptible fraction x^- from $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$.

The quantity that is conserved along epidemic trajectories is a function of susceptible fraction and epidemic momentum, and can be written [5, Section 3(d)]

$$C(x, y) = y + \hat{x} V\left(\frac{x}{\hat{x}}\right), \quad (2.8)$$

where $V(u) = u - 1 - \ln u$ is the ‘‘Volterra function’’ [16] and $\hat{x} = \frac{1}{\mathcal{R}_0}$. The conserved value of C is the maximum epidemic momentum,

$$C(X(\tau), Y(\tau)) = C(\hat{x}, \hat{y}) = \hat{y}. \quad (2.9)$$

At the pre- and post-epidemic endpoints, $X = x^\pm$ and $Y = 0$. Consequently, $\hat{y} = C(x^\pm, 0) = \frac{1}{\mathcal{R}_0} V(\mathcal{R}_0 x^\pm)$. Since Eq. (2.3) gives $\mathcal{R}_0 x^\pm = 1/\mathcal{L}_\pm$, we obtain

$$\mathcal{R}_0 = \frac{1}{\hat{y}} V\left(\frac{1}{\mathcal{L}_\pm}\right). \quad (2.10)$$

Thus either epidemic tail provides an exact expression for $\mathcal{R}_0$ in terms of its tail exponent, λ^- or λ^+ , the generation interval distribution $g(\alpha)$, and the maximum epidemic momentum $\hat{y}$, with no dependence on the pre-epidemic susceptible fraction x^- .

For inference from the initial growth phase, we use $\mathcal{L}_\pm = \mathcal{L}_-$ in Eq. (2.10). Then Eq. (2.3) gives

$$x^- = \frac{1}{\mathcal{R}_0 \mathcal{L}_-} \quad \text{and} \quad z^- = 1 - x^- = 1 - \frac{1}{\mathcal{R}_0 \mathcal{L}_-}. \quad (2.11)$$

Hence estimates of λ^- and $\hat{y}$, together with $g(\alpha)$, determine both pathogen transmissibility ($\mathcal{R}_0$) and prior population immunity (z^-).

The falling tail exponent λ^+ is difficult to estimate because the asymptotic decay occurs when incidence is very low. However, it is not necessary to observe the falling tail in order

to estimate the final size of the epidemic. Applying Eq. (2.10) to the two endpoints gives

$$V\left(\frac{1}{\mathcal{L}_-}\right) = V\left(\frac{1}{\mathcal{L}_+}\right). \quad (2.12)$$

The solution corresponding to the post-epidemic endpoint is

$$\frac{1}{\mathcal{L}_+} = -W_+\left(-\frac{1}{\mathcal{L}_-}e^{-1/\mathcal{L}_-}\right), \quad (2.13)$$

where W_+ is the principal branch of Lambert’s W function (Section C). Since $x^+ = 1/(\mathcal{R}_0\mathcal{L}_+)$, the proportion infected during the focal epidemic—the final size—is therefore

$$z^+ = x^- - x^+ = \frac{1}{\mathcal{R}_0} \left(\frac{1}{\mathcal{L}_-} - \frac{1}{\mathcal{L}_+} \right). \quad (2.14)$$

Operationally, we estimate λ^- from the initial exponential-growth phase, evaluate $\mathcal{L}_-$, construct $Y(\tau)$ from the incidence history, and identify its maximum $\hat{y}$. Equations (2.10) and (2.11) determine $\mathcal{R}_0$, x^- , and z^- . Equation (2.13) then determines $\mathcal{L}_+$, and Eq. (2.14) gives z^+ ; the post-epidemic susceptible fraction follows from $x^+ = x^- - z^+$. Observations must extend long enough to estimate the maximum of epidemic momentum, but need not extend into the asymptotic falling tail.

The absolute scale of the incidence curve is required because $\hat{y}$, unlike λ^- and $\mathcal{L}_-$, depends on that scale. Multiplying incidence by a constant multiplies $Y(\tau)$ and $\hat{y}$ by the same constant, and therefore changes the value of $\mathcal{R}_0$ inferred from Eq. (2.10). Incidence must therefore be expressed as a fraction of the population, or the factor connecting the observed time series to incidence must be known or examined explicitly, as in the influenza mortality application below.

3 Results

3.1 Estimates of prior immunity and $\mathcal{R}_0$ from stochastic simulations

We assessed the momentum-based method using stochastic SEIR simulations for which $\mathcal{R}_0$ and the pre-epidemic susceptible fraction x^- were known. The simulations spanned a wide range of $\mathcal{R}_0$ for several combinations of x^- and population size N , allowing us to examine the effects of demographic stochasticity on each stage of the inference.

The method first estimates the initial growth rate λ^- from the rising tail and constructs the epidemic momentum from incidence to estimate its maximum $\hat{y}$. (Details of the computation, including estimating initial conditions, are given in Section A.) Figure 1 shows the relative errors in these two quantities. Inserting the estimated values of λ^- and $\hat{y}$ into Eqs. (2.10) and (2.11) gives the corresponding estimates of $\mathcal{R}_0$ and x^- , shown in Fig. 2.

In the upper panels of Fig. 2, grey squares show the true values of x^- . The estimated values are distributed around the corresponding true values across the simulated parameter combinations, with greater stochastic variation in smaller populations. The lower panels

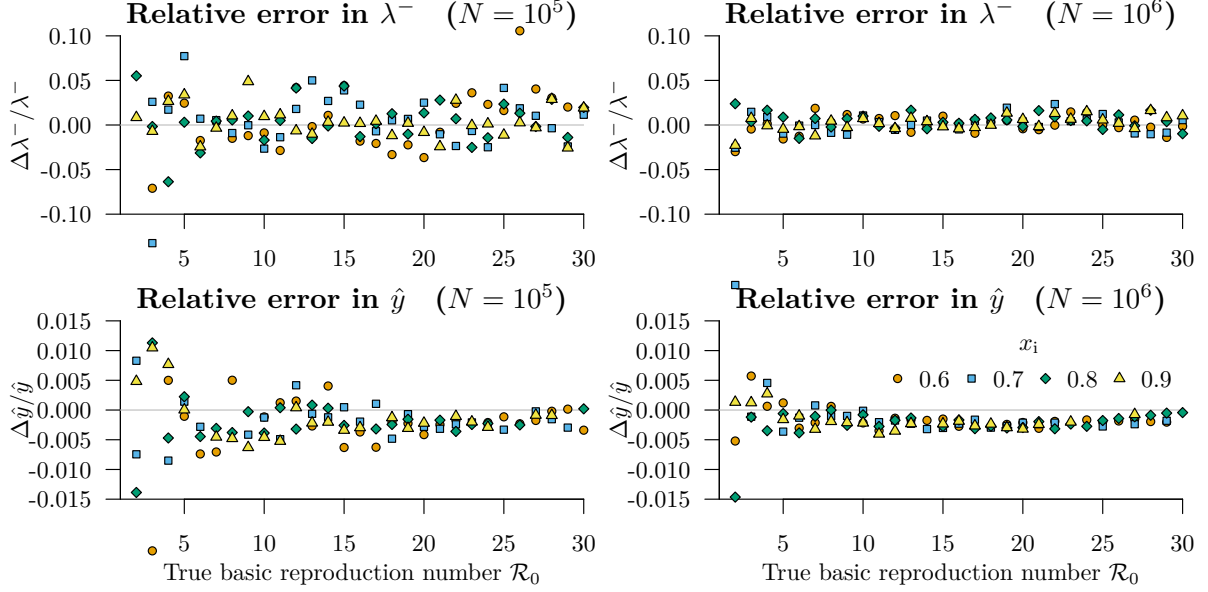


Figure 1: **Estimates of initial growth rate λ^- and peak epidemic momentum $\hat{y}$ from stochastic SEIR simulations.** The top panels show the relative error in the initial growth rate λ^- as a function of the true $\mathcal{R}_0$ specified in the simulations (with population size $N = 10^5$ on the left and $N = 10^6$ on the right). The exact value of λ^- is given by the SEIR row in Table 1. Simulations were carried out with equal mean latent and infectious periods ($\ell = 1$) and incidence time series were obtained by “observing” five times per infectious period, corresponding to daily data for a disease with a five day infectious period. Realistic initial conditions were chosen by ensuring approximate agreement with the initial exponential growth phase of the SEIR ODEs. Estimates of λ^- were obtained by applying the R package `epigrowthfit` [2, 17, 18] to the simulated incidence time series. The second row of panels shows the relative error in the peak epidemic momentum ($\hat{y}$, the exact value of which is given by $Y(\hat{x})$ in [5, Eq. (2.4)]); the epidemic momentum $Y(\tau)$ was estimated by convolving the simulated cumulative incidence $\bar{\iota}(\tau)$ with the generation interval distribution $g(\alpha)$ [Eq. (2.6) and Table 1]. A small systematic underestimate in $\hat{y}$ is evident, but the magnitude of the relative error in $\hat{y}$ is an order of magnitude smaller than the magnitude of the relative error in λ^- , so the systematic error has a negligible effect on the estimate of $\mathcal{R}_0$.

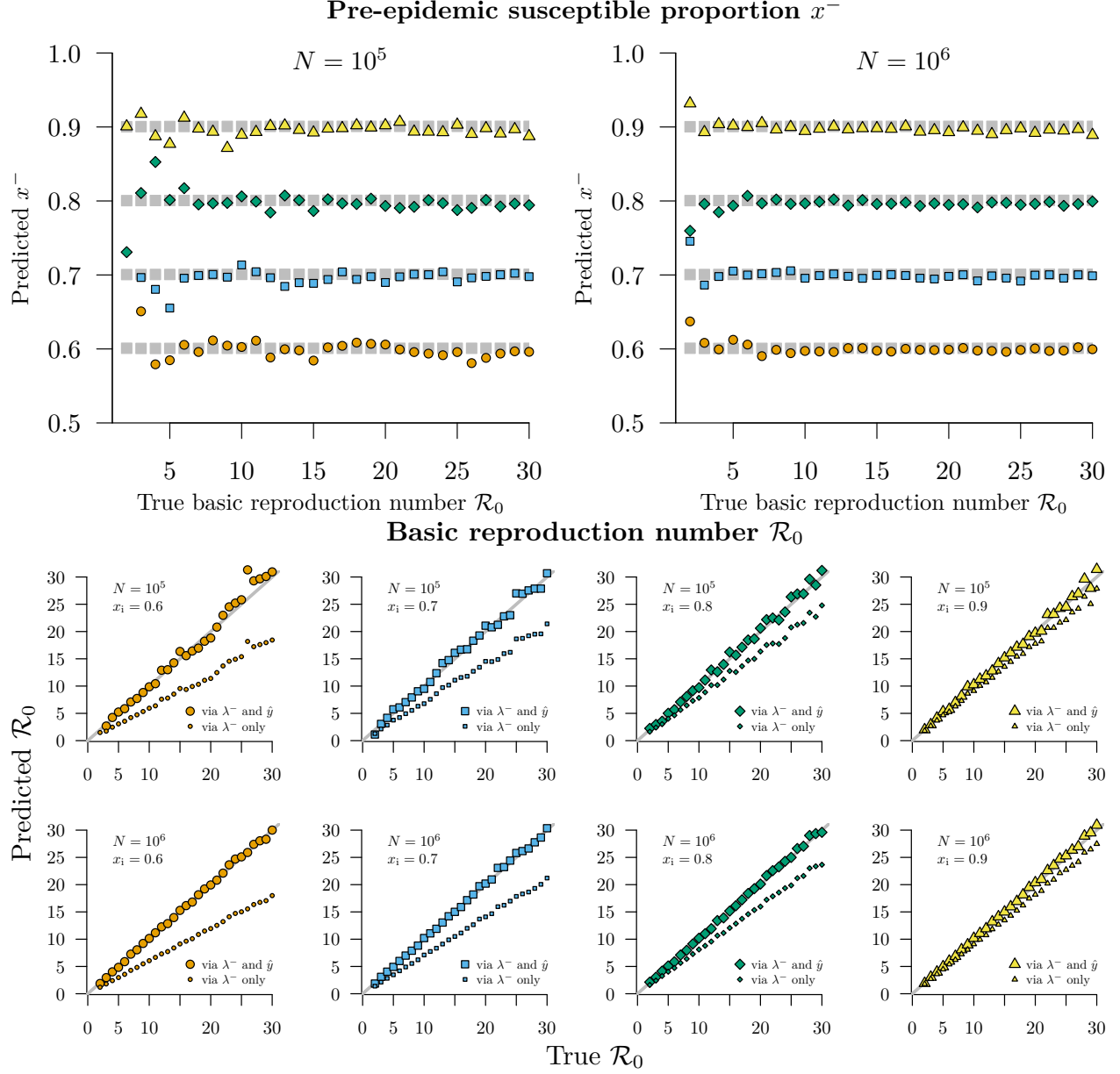


Figure 2: **Prior population immunity ($z^- = 1 - x^-$) and basic reproduction number ($\mathcal{R}_0$) estimated from stochastic SEIR simulations.** Exploiting the epidemic momentum, we successfully disentangle and accurately estimate both z^- and $\mathcal{R}_0$. The top panels show the predicted pre-epidemic susceptible proportion x^- , so the pre-existing level of population immunity is $z^- = 1 - x^-$ [estimated via Eq. (2.11)]. The true x^- associated with the deterministic skeleton of the model [computed via Eq. (2.11) using the exact value of λ from Table 1] is indicated with grey squares. Symbols and colours are associated with the initial susceptible proportion x_i as in Fig. 1. The bottom panels show the predicted $\mathcal{R}_0$ from the same simulations. The smaller symbols show the value of $\mathcal{R}_0$ estimated using the uncorrected Wallinga-Lipsitch (WL) formula [1], which uses only the estimated growth rate λ , whereas the larger symbols show $\mathcal{R}_0$ as estimated using Eq. (2.10), which uses both λ and the estimated peak epidemic momentum $\hat{y}$. The grey line corresponds to “Predicted $\mathcal{R}_0$ = True $\mathcal{R}_0$ ”.

show the estimates of $\mathcal{R}_0$, which similarly lie close to the line of perfect agreement, with accuracy improving as population size increases.

For comparison, the smaller symbols in the lower panels show the reproduction numbers obtained from the initial growth rate under the assumption that the population was initially fully susceptible ($x^- = 1$). When pre-existing immunity is present, these values are systematically smaller than the true $\mathcal{R}_0$, because the initial growth rate identifies $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$ rather than $\mathcal{R}_0$.

All simulations shown in Figs. 1 and 2 had equal mean latent and infectious periods ($\ell = 1$ in Eq. (T6)); the performance of the method was similar for other values of ℓ .

3.2 Reappraisal of the 1918 influenza pandemic in Philadelphia

Having shown that the momentum-based method recovers known values of $\mathcal{R}_0$ and x^- from stochastic simulations, we applied it to daily pneumonia and influenza (P&I) mortality recorded during the main wave of the 1918 influenza pandemic in Philadelphia, which occurred in the autumn of 1918 (Fig. 3).

Because mortality rather than incidence was reported, we used Richardson–Lucy deconvolution to reconstruct a curve proportional to incidence [6]. (Details of the mortality data, the infection-to-death distribution used for deconvolution, the generation-interval assumptions, and the calculation of epidemic momentum are given in the caption to Fig. 3.) The proportionality factor includes the case-fatality proportion, **CFP**, because only that fraction of infections resulted in recorded deaths. This unknown scale does not affect the estimated initial growth rate [2, 17, 18], which was $\lambda^- \simeq 0.16/\text{day}$.

Following Mills *et al.* [24], we assumed mean latent and infectious periods of $T_{\text{lat}} = 1.9$ days and $T_{\text{inf}} = 4.1$ days, respectively, hence $\ell \equiv T_{\text{lat}}/T_{\text{inf}} = 0.463$. Measuring time in units of T_{inf} gives $\lambda^- \approx 0.655$. The corresponding SEIR generation-interval distribution [Table 1] then yields

$$\mathcal{L} \approx 0.464. \quad (3.1)$$

This estimate gives a lower bound on $\mathcal{R}_0$ that is independent of the incidence scale: since $x^- \leq 1$, we have $\mathcal{R}_0 \geq \mathcal{R}_{\text{eff}}$, so Eq. (2.3) implies

$$\mathcal{R}_0 \geq \frac{1}{\mathcal{L}} \approx 2.16. \quad (3.2)$$

Convolving the reconstructed incidence curve with the remaining reproductive-potential kernel [Eq. (2.4)] gives the epidemic momentum [Eq. (2.5)] shown in Fig. 3. Because epidemic momentum is linearly related to incidence, the multiplicative factor **CFP** is retained in the reconstructed momentum curve; its maximum is therefore $\text{CFP} \times \hat{y}$, rather than $\hat{y}$. For mortality data, Eq. (2.10) is therefore more usefully written

$$\mathcal{R}_0 = \frac{\text{CFP}}{\text{CFP} \times \hat{y}} V\left(\frac{1}{\mathcal{L}}\right). \quad (3.3)$$

The momentum curve in Fig. 3 gives

$$\text{CFP} \times \hat{y} \approx 0.00287, \quad (3.4)$$

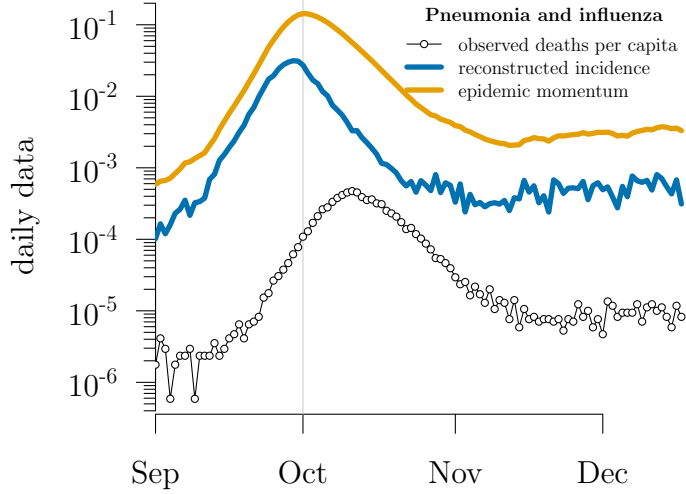


Figure 3: **1918 influenza pandemic in Philadelphia, USA.** Daily deaths from pneumonia and influenza (P&I) were recorded from 1 September to 31 December 1918 [19]. We deconvolved the observed mortality time series to obtain estimated daily incidence $\iota(t)$, using an empirically estimated infection to death distribution: as detailed in previous work [6] and implemented in the `fastbeta` R package [20, 21], gamma distributions were fitted to an empirical incubation period distribution [22, Figure 1] and an empirical symptom onset to death distribution [23, Chart 2], which were then convolved to obtain the infection to death distribution. We then convolved the estimated $\iota(t)$ with the estimated reduced reproduction number [Eq. (2.4)] (via $g(\alpha)$ from Table 1 with $\ell = 0.463$) to obtain the epidemic momentum time series $Y(t)$ [Eq. (2.5)]. The peak of the observed daily P&I mortality occurred on 11 October 1918 (with 803 P&I deaths), whereas estimated incidence peaked on 29 September 1918 and estimated epidemic momentum peaked on 1 October 1918 (vertical grey line). Note that peak momentum always occurs after peak incidence (see Section B). Associated estimates of $\mathcal{R}_0$ and population immunity are discussed in the main text in Section 3.2.

so that

$$\mathcal{R}_0 \approx 135 \times \text{CFP}. \quad (3.5)$$

The lower bound on $\mathcal{R}_0$ given in Eq. (3.2) therefore implies a CFP lower bound,

$$\text{CFP} \approx \frac{\mathcal{R}_0}{135} \gtrsim \frac{2.16}{135} = 1.6\%. \quad (3.6)$$

Frost [25, p. 593] estimated that the case-fatality proportion during the main wave of the 1918 pandemic ranged from 0.8% to 3.1% across US cities. For a group of northeastern communities near Philadelphia, he estimated $\text{CFP} = 2.05\%$ [25, p. 593], consistent with the lower bound in Eq. (3.6). We therefore use $\text{CFP} \simeq 2\%$ as a representative value for Philadelphia. Substitution into Eq. (3.5) implies

$$\mathcal{R}_0 \approx 2.71. \quad (3.7)$$

Equation (2.11) then gives

$$z^- \approx 0.203, \quad (3.8)$$

indicating that approximately 20% of the population was immune before the main wave.

Mills *et al.* [24] also assumed $\text{CFP} = 2\%$, based on the midpoint of Frost’s range [25], but assumed prior population immunity $z^- = 30\%$ from seasonal-influenza evidence [24, pp.905–906] (the relevant passages are reproduced in Section D). They reported

$$1.7 \leq \mathcal{R}_0 \leq 2.4 \quad (3.9)$$

for Philadelphia [24, SI p.5]. However, using the observed initial growth rate and the SEIR generation-interval distribution assumed both here and by Mills *et al.*, $z^- = 0.3$ would imply

$$\mathcal{R}_0 = \frac{1}{(1 - z^-)\mathcal{L}} \approx 3.08, \quad (3.10)$$

which lies outside their reported range. Thus their stated assumption about prior immunity and their reported estimate of $\mathcal{R}_0$ are not mutually consistent under Eq. (2.11).

Our momentum-based analysis instead estimates $\mathcal{R}_0$ and z^- consistently from the observed initial growth and reconstructed momentum curve. The estimate in Eq. (3.8) suggests that a substantial fraction (20%) of the Philadelphia population was already immune before the main wave, plausibly because of infection during the spring 1918 “herald wave”.

4 Discussion

Initial epidemic growth does not, by itself, distinguish pathogen transmissibility from pre-existing population immunity. Rather, it identifies the effective reproduction number $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$, the product of the basic reproduction number and the fraction of the population susceptible before the epidemic began [4]. Consequently, different combinations of $\mathcal{R}_0$ and x^- can generate the same initial growth. We have shown that epidemic momentum provides the additional information needed to separate these quantities. Together, the initial growth rate (tail exponent) and maximum epidemic momentum determine $\mathcal{R}_0$ and x^- , from which prior immunity (z^-) and the proportion infected during the focal epidemic (z^+) follow.

The momentum-based method successfully recovered both $\mathcal{R}_0$ and x^- from stochastic SEIR simulations across the parameter combinations examined (Figs. 1 and 2). As expected, stochastic variation was greater in smaller populations, but the estimates were distributed around the corresponding true values. By contrast, estimates obtained from the initial growth rate while assuming $x^- = 1$ were systematically smaller than the true $\mathcal{R}_0$ whenever pre-existing immunity was present. These estimates are not intrinsically inaccurate: they estimate $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 x^-$, rather than $\mathcal{R}_0$. The error arises when $\mathcal{R}_{\text{eff}}$ is interpreted as the basic reproduction number despite uncertainty about the initial susceptible fraction.

Our reappraisal of the 1918 influenza pandemic illustrates the biological importance of this distinction. For the main wave in Philadelphia, which occurred in the autumn of 1918, the observed initial growth and reconstructed epidemic momentum imply $\mathcal{R}_0 \approx 2.71$ and $z^- \approx 0.203$. Thus, approximately 20% of the population appears to have been immune before the main wave began. Infection during the spring 1918 “herald wave” provides a plausible explanation for at least some of this immunity, but other sources of pre-existing immunity are also possible.

The Philadelphia analysis also illustrates the consequences of assuming prior immunity rather than estimating it. Mills *et al.* [24] assumed $z^- = 30\%$, but the observed growth rate

and their assumed generation-interval distribution imply $\mathcal{R}_0 \approx 3.08$, outside the range they reported. Thus, their assumed prior immunity and reported estimate of $\mathcal{R}_0$ are not mutually consistent under the standard relationship between epidemic growth, prior immunity, and reproduction number given in Eq. (2.11). The broader point is not specific to this earlier analysis: estimates of $\mathcal{R}_0$ obtained by fixing prior immunity necessarily inherit any error in that assumption.

4.1 Relation to previous approaches

Previous attempts to infer transmissibility and prior immunity from epidemic time series have generally required additional approximations, strong model assumptions, or external information. Caley *et al.* [26] estimated the time-dependent effective reproduction number during the multi-wave 1919 influenza epidemic in Sydney from hospitalization and mortality data. They combined these estimates with cumulative incidence and *a priori* assumptions about periods in which contact behaviour had returned to normal to separate susceptible depletion from changes in contact rates. Their analysis suggested that more than 90% of the population was initially susceptible and gave a preferred estimate $\mathcal{R}_0 \approx 1.8$. The inference of prior immunity depended on the assumed clinical attack rate, the reconstruction of the time-dependent reproduction number, and the correct identification of periods without behavioural change.

Cope *et al.* [27] used approximate Bayesian computation to fit a climate-driven stochastic epidemic model to seasonal influenza surveillance data from New South Wales. Their posterior distribution showed a strong relationship between $\mathcal{R}_0$ and the initial susceptible fraction: relatively high values of $\mathcal{R}_0$ were associated with low susceptibility, while their product was much more tightly constrained. Thus, even with a substantially richer model and observation process, the data did not sharply identify the two quantities separately.

Bergström *et al.* [28] proved that, in an SIR model with under-reporting and prior immunity, the transmission rate, reporting fraction, and initial immune fraction are not jointly identifiable from reported incidence alone. They showed that identifiability can be restored by supplementing reported incidence with an estimate of prior immunity or infection prevalence. Their analysis is restricted to the SIR model and assumes the final size of the epidemic has been observed. Our approach, which applies to all models represented by the renewal equation, instead exploits the fact that epidemic momentum peaks at $\hat{x} = \frac{1}{\mathcal{R}_0}$ so all required information—including an estimate of peak momentum $\hat{y}$ —can be inferred much earlier. However, an unknown reporting fraction still determines the scale of epidemic momentum and must be estimated or constrained; in the Philadelphia analysis, the corresponding unknown scale is the case-fatality proportion.

4.2 Data requirements and statistical challenges

The momentum-based method requires an estimate of the intrinsic generation-interval distribution. This requirement is not unique to our method: the same distribution is already needed to convert an epidemic growth rate into an effective reproduction number [1]. Nevertheless, uncertainty or misspecification in $g(\alpha)$ affects both $\mathcal{R}_{\text{eff}}$ and the reconstruction of epidemic momentum, and therefore propagates into estimates of $\mathcal{R}_0$ and x^- . Applications

should therefore be mindful of sensitivity to the assumed form and parameters of the intrinsic generation-interval distribution.

The absolute scale of incidence is also required for point estimates of $\mathcal{R}_0$ and x^- . Multiplying the incidence curve by a constant leaves its exponential growth rate unchanged but multiplies epidemic momentum and its maximum by the same constant, thereby changing the inferred $\mathcal{R}_0$. Incidence must therefore be expressed as a fraction of the population, or the factor connecting the observed time series to incidence must be known or estimated. Even when this scale factor cannot be estimated reliably, the initial growth rate still provides a useful lower bound on $\mathcal{R}_0$. For Philadelphia, Eq. (3.2) gives $\mathcal{R}_0 \gtrsim 2.16$, independently of the assumed case-fatality proportion.

Mortality, hospital admissions, and other delayed observations require reconstruction of the underlying incidence curve. This reconstruction depends on the relevant delay distribution and may introduce additional uncertainty. The unobserved incidence history before observations begin must also be approximated, although the initial exponential phase provides a natural extrapolation [Section A].

Peak epidemic momentum occurs after peak incidence [Section B], but this does not imply that the entire rise to $\hat{y}$ must be observed before a useful estimate can be made. As an epidemic unfolds, provisional estimates of the eventual maximum $\hat{y}$, and hence of $\mathcal{R}_0$ and x^- , should become progressively more informative. Moreover, when incidence is reconstructed from a delayed observable such as hospitalization or mortality, a useful estimate of peak momentum may be available well before the observed time series reaches its peak, as illustrated by Fig. 3. An important statistical challenge will be to develop robust confidence intervals for $\mathcal{R}_0$ and x^- and to determine how they improve as more of an epidemic is observed.

4.3 Model assumptions and extensions

The derivations in this paper assume mass-action incidence, so that susceptibility enters incidence linearly through the susceptible fraction of the population. Individual differences in intrinsic susceptibility, contact rate, and other mechanisms can instead produce nonlinear dependence of incidence on the susceptible fraction [29, 30, 31, 32]. In that case, the conserved quantity and inference formulae derived here must be modified.

We have extended the epidemic-momentum framework to incidence that depends nonlinearly on susceptible fraction, and derived the corresponding conserved quantity in this more general setting [33]. This class provides one way to incorporate nonlinear effects of susceptibility without explicitly tracking population structure [32]. More detailed forms of host heterogeneity may instead require structured models that distinguish among host types.

The present analysis also assumes that the underlying transmissibility and intrinsic generation-interval distribution remain fixed over the period analysed. Behavioural change, interventions, pathogen evolution, or changes in reporting can violate these assumptions and break the conservation law [5, Appendix J]. Applications should therefore be restricted to periods over which these assumptions are defensible, or would require the development of extensions that account explicitly for time-varying transmission.

Finally, the method estimates the susceptible fraction relevant to the focal epidemic but does not identify the biological source of prior immunity. Such immunity may arise from previous infection with the same pathogen, cross-reactive immunity to related strains,

vaccination, or other differences in susceptibility. In the Philadelphia application, infection during the spring herald wave is one plausible source, but other sources of pre-existing immunity are possible.

4.4 Conclusions

Epidemic growth reflects both pathogen transmissibility and the immunological state of the population in which transmission occurs. Assuming that the population was initially fully susceptible can therefore lead to substantial underestimation of $\mathcal{R}_0$ and misinterpretation of epidemic final size. Epidemic momentum provides a way to separate these effects using information contained in the same epidemic time series. By combining the initial growth rate with the maximum epidemic momentum, the momentum-based method yields separate estimates of $\mathcal{R}_0$ and prior population immunity while retaining the generality of the renewal-equation framework.

Ethics

This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility

The 1918 pneumonia and influenza mortality data shown in Fig. 3 have been published previously [6] and are available in the `fastbeta` R package [21]. R code that reproduces all the figures will be included as supplementary material to the published version.

Declaration of AI use

We used ChatGPT and OpenAI Codex to assist with R coding, L^AT_EX coding, and language editing. All scientific arguments, mathematical derivations, analyses, interpretations, and conclusions were developed by the authors, who take full responsibility for the final manuscript.

Author contributions

Both authors contributed to all aspects of the work.

Competing interests

We declare we have no competing interests.

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APPENDICES

A Epidemic momentum from finite incidence history

For $\tau \geq \tau_i$, splitting Eq. (2.5) over infections that occurred before and after observations began gives

$$Y(\tau) = \int_{\tau-\tau_i}^{\infty} \iota(\tau-\alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha + \int_0^{\tau-\tau_i} \iota(\tau-\alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (\text{A1})$$

The second integral is computed directly from the observed incidence history. During the initial exponential-growth phase, Eq. (2.2), with $\iota(\tau_i) = \iota_i$, gives

$$\iota(\tau-\alpha) \approx \iota_i e^{\lambda(\tau-\tau_i)} e^{-\lambda\alpha}, \quad \alpha > \tau - \tau_i, \quad (\text{A2})$$

where the initial growth rate λ is also estimated from the observed incidence time series [2, 17, 18]. The first integral in Eq. (A1) is therefore approximated by

$$\iota_i e^{\lambda(\tau-\tau_i)} \int_{\tau-\tau_i}^{\infty} e^{-\lambda\alpha} \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (\text{A3})$$

Using the definition of $\mathcal{R}_\alpha$ in Eq. (2.4), we obtain

$$\begin{aligned} \int_{\tau-\tau_i}^{\infty} e^{-\lambda\alpha} \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha &= \int_0^{\infty} e^{-\lambda(\alpha+\tau-\tau_i)} \int_{\alpha+\tau-\tau_i}^{\infty} g(\alpha') d\alpha' d\alpha \\ &= e^{-\lambda(\tau-\tau_i)} \int_0^{\infty} e^{-\lambda\alpha} \int_{\alpha}^{\infty} g(\alpha' + \tau - \tau_i) d\alpha' d\alpha \\ &= e^{-\lambda(\tau-\tau_i)} \int_0^{\infty} \int_0^{\alpha'} e^{-\lambda\alpha} d\alpha g(\alpha' + \tau - \tau_i) d\alpha' \\ &= \frac{e^{-\lambda(\tau-\tau_i)}}{\lambda} \int_0^{\infty} (1 - e^{-\lambda\alpha'}) g(\alpha' + \tau - \tau_i) d\alpha' \\ &= \frac{e^{-\lambda(\tau-\tau_i)}}{\lambda} [\mathcal{L}[g_{\tau-\tau_i}](0) - \mathcal{L}[g_{\tau-\tau_i}](\lambda)], \end{aligned} \quad (\text{A4})$$

where

$$g_{\tau-\tau_i}(\alpha) = g(\alpha + \tau - \tau_i) \quad (\text{A5})$$

is the shifted generation-interval kernel, with Laplace transform

$$\mathcal{L}[g_{\tau-\tau_i}](\lambda) = \int_0^\infty e^{-\lambda\alpha} g(\alpha + \tau - \tau_i) d\alpha. \quad (\text{A6})$$

Substituting Eq. (A4) into Eqs. (A1) and (A3) gives the finite-history representation

$$Y(\tau) \approx \frac{\iota_i}{\lambda^-} [\mathcal{L}[g_{\tau-\tau_i}](0) - \mathcal{L}[g_{\tau-\tau_i}](\lambda^-)] + \int_0^{\tau-\tau_i} \iota(\tau-\alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (\text{A7})$$

The first term reconstructs the contribution from infections that occurred before observations began, while the second is calculated directly from the observed incidence history.

For the SEIR generation-interval distribution given in Table 1, the shifted Laplace transform is

$$\mathcal{L}[g_{\tau-\tau_i}](\lambda) = \begin{cases} \frac{1}{1-\ell} \left[\frac{e^{-(\tau-\tau_i)}}{1+\lambda} - \frac{\ell e^{-(\tau-\tau_i)/\ell}}{1+\ell\lambda} \right], & \ell \neq 1, \\ \frac{1 + (1+\lambda)(\tau-\tau_i)}{(1+\lambda)^2} e^{-(\tau-\tau_i)}, & \ell = 1. \end{cases} \quad (\text{A8})$$

It follows that the contribution from the unobserved incidence history is

$$\frac{\iota_i}{\lambda^-} [\mathcal{L}[g_{\tau-\tau_i}](0) - \mathcal{L}[g_{\tau-\tau_i}](\lambda^-)] = \begin{cases} \frac{\iota_i}{1-\ell} \left[\frac{e^{-(\tau-\tau_i)}}{1+\lambda^-} - \frac{\ell^2 e^{-(\tau-\tau_i)/\ell}}{1+\ell\lambda^-} \right], & \ell \neq 1, \\ \iota_i e^{-(\tau-\tau_i)} \frac{2 + (\tau-\tau_i) + \lambda^- [1 + (\tau-\tau_i)]}{(1+\lambda^-)^2}, & \ell = 1. \end{cases} \quad (\text{A9})$$

These expressions allow the contribution from infections that occurred before τ_i to be calculated directly in the SEIR examples, without numerical integration over an unobserved incidence history.

B Incidence always peaks before momentum

Differentiating under the integral sign in Eq. (2.5), we have

$$\frac{dY}{d\tau} = \int_0^\infty \frac{d\iota}{d\tau}(\tau-\alpha) \frac{\mathcal{R}_\alpha}{\mathcal{R}_0} d\alpha. \quad (\text{B1})$$

We must have $\frac{d\iota}{d\tau} > 0$ until incidence reaches its peak (where $\frac{d\iota}{d\tau} = 0$), and thus $\frac{dY}{d\tau} > 0$ at peak incidence; therefore, momentum always peaks after incidence. Peak momentum ($\hat{y}$) always occurs at $\hat{x}$ [5, Section 3(a)], whereas Eq. (2.1a) shows that the fraction susceptible is monotone decreasing. Hence the fraction susceptible at peak incidence always exceeds $\hat{x}$.

C Lambert’s W -function

This appendix is identical to the corresponding appendix in [5, Appendix F] and is included here for convenience.

If $\mathcal{E}(z) = ze^z$, Lambert’s W -function $W(z)$ ([34]; [35, §4.13]) solves the “left-sided” inverse relation $\mathcal{E}(W(z)) = z$. This equation has countably many solutions, written $W_k(z)$ for solutions with $\arg z \in [2\pi k, 2\pi(k+1))$. Only W_0 and W_{-1} return real values for real z ; for other k , W_k is always complex. We use the two real branches: W_{-1} maps $[-\frac{1}{e}, 0)$ to $(-\infty, -1]$, and W_0 maps $[-\frac{1}{e}, \infty)$ to $[-1, \infty)$. For these two branches, W_k is a *partial* “right-sided” inverse function for $\mathcal{E}(z)$:

$$W_{-1}(\mathcal{E}(z)) = z \quad \text{if } z \leq -1 \quad (\text{C1a})$$

$$W_0(\mathcal{E}(z)) = z \quad \text{if } z \geq -1. \quad (\text{C1b})$$

While the standard notation W_k is chosen to indicate the winding number associated with the given branch, for our purposes it is more convenient to write W_- for W_{-1} and W_+ for W_0 , so we can write expressions involving $W_\pm$, where the $\pm$ matches the corresponding sign in $x^\pm$ and/or $\lambda^\pm$ (W_+ and W_- are also written W_p and W_m [35, §4.13]).

D Previous estimates of prior immunity to 1918 influenza

Mills *et al.* [24, pp. 905–906] state: “The proportion of the population susceptible at the start of the pandemic determines the relationship between R and the basic reproductive number ($\mathcal{R}_0$), which is the number of secondary cases generated by a primary case in a completely susceptible population². Frost hypothesized that a 1918 pandemic-like strain spread throughout America in the spring of 1918 (ref. 22), and recent analyses support this ‘herald wave’ hypothesis²³. Anecdotal evidence suggests that those who fell ill in the spring were protected from disease in the autumn pandemic²⁴. Nevertheless, a large majority of the population was probably susceptible to the A/H1N1 pandemic strain in September 1918. In a typical epidemic transmission season, 15–25% of the population becomes infected with influenza⁴. The herald wave is believed to have arrived late in the 1917–18 transmission season. Using 70% as a conservative lower bound for the fraction susceptible at the start of the autumn pandemic, the medians for our initial and extreme $\mathcal{R}_0$ are 2.9 and 3.9.”

Frost [25, p. 593] states: “The case fatality, or ratio of deaths to total cases of influenza, varied in the localities surveyed from 3.1 per cent in New London to 0.8 per cent in San Antonio, the variations showing no consistent relation to incidence rates. There is, however, some relation to geographic location, namely, that the highest case-fatality rates occurred in New London, San Francisco, Baltimore, and minor Maryland communities, in the order named—that is, in communities representing, respectively, the northern half of the Atlantic seaboard and the Pacific coast. In the central and southern cities the case fatality was generally notably lower. Combining the eleven localities into three groups comprising, respectively—(1) San Francisco, (2) New London, Baltimore, and minor Maryland communities,

(3) central and southern cities, comprising all other localities, the case-fatality rates in these three groups are, respectively, 2.33, 2.05, and 1.08 per cent. This is of interest in connection with the observation that from the standpoint of mortality rates the epidemic was generally more severe along the northern Atlantic Seaboard and the Pacific Coast than in the Central States.”

Table 1: **Standard infectious disease transmission models.**

The susceptible-infectious-removed (**SIR**) model, first proposed by KM [7], assumes that all infected individuals are equally infectious, and immunity upon recovery is permanent. It is represented with two equations in standard form [(T1), with parameters β , the transmission rate, γ , the removal rate, and population size N] or dimensionless form [(T2), with parameter $\mathcal{R}_0$, the basic reproduction number, and time measured in units of the mean infectious period, $T_{\text{inf}} = \gamma^{-1}$]. The generation interval distribution is identical to the infectious period distribution, so the mean generation interval is $\mu = \gamma^{-1}$. In this simple model, the epidemic momentum is equal to the prevalence.

Most infectious diseases have a non-negligible *latent period*, *i.e.*, there is a delay between initial infection and becoming infectious. The susceptible-exposed-infectious-removed (**SEIR**) model introduces an *exposed* stage (E) of mean duration γ_E^{-1} , when individuals are infected but not yet infectious [36]. The mean generation interval μ is the sum of the means of the latent and infectious periods [10, 11]. In dimensionless units, we write the mean latent period ℓ , *i.e.*, as a proportion of the mean infectious period, so the mean generation interval is $\mu = \ell + 1$ in these units. The standard form is (T5) and the dimensionless form is (T6). We denote the proportions susceptible, exposed, and infectious by X , Y_E , and Y_I , respectively, and—as in the SIR model—the epidemic momentum Y corresponds to the *total proportion infected*, *i.e.*, $Y = Y_E + Y_I$ (see [5, Section 2(c)]), consistent with our notation for the SIR model (T2). The *per capita* rates at which individuals leave the exposed and infectious compartments are γ_E and γ_I , respectively. The basic reproduction number is $\mathcal{R}_0 = \beta/\gamma_I$ and the mean latent period (as a proportion of the mean infectious period γ_I^{-1}) is $\ell = \gamma_I/\gamma_E$.

Generic epidemic models can be specified using the **renewal equation**, which relates the susceptible fraction X to the force of infection F with a differential equation (T9a), and relates F to the generation interval distribution, $g(\alpha)$, via a convolution [Eq. (T9b)]. If $g(\alpha)$ is not known, it is common to assume it is a gamma distribution, as in (T11).

SIR model

Standard	Dimensionless ($X = \frac{S}{N}, Y = \frac{I}{N}$)	Parameters	Properties
		$\beta =$ transmission rate	
$\frac{dS}{dt} = -\frac{\beta}{N} S I$ (T1a)	$\frac{dX}{d\tau} = -\mathcal{R}_0 X Y$ (T2a)	$\frac{1}{\gamma} =$ mean infectious period	$\tau = \gamma t$ (T4a)
$\frac{dI}{dt} = \left(\frac{\beta}{N} S - \gamma\right) I$ (T1b)	$\frac{dY}{d\tau} = (\mathcal{R}_0 X - 1) Y$ (T2b)	$\mu =$ mean generation interval $= \frac{1}{\gamma}$ (T3c)	$g(\alpha) = e^{-\alpha}$ (T4b)
	$\iota = \mathcal{R}_0 X Y$ (T2c)	$\mathcal{R}_0 =$ basic reproduction number $= \frac{\beta}{\gamma}$ (T3d)	$\mathcal{L}[g](\lambda) = \frac{1}{\lambda+1}$ (T4c)
			$\lambda^\pm = \mathcal{R}_0 x^\pm - 1$ (T4d)

SEIR model

Standard	Dimensionless ($Y_E = \frac{E}{N}, Y_I = \frac{I}{N}$)	Parameters	Properties
		$\frac{1}{\gamma_E} =$ mean latent period	$\tau = \gamma_I t$ (T8a)
$\frac{dS}{dt} = -\frac{\beta}{N} S I$ (T5a)	$\frac{dX}{d\tau} = -\mathcal{R}_0 X Y_I$ (T6a)	$\frac{1}{\gamma_I} =$ mean infectious period	$g(\alpha) = \begin{cases} \alpha e^{-\alpha}, & \ell = 1 \\ \frac{e^{-\alpha} - e^{-\alpha/\ell}}{1-\ell}, & \ell \neq 1 \end{cases}$ (T8b)
$\frac{dE}{dt} = \frac{\beta}{N} S I - \gamma_E E$ (T5b)	$\frac{dY_E}{d\tau} = \mathcal{R}_0 X Y_I - \frac{1}{\ell} Y_E$ (T6b)	$\mu = \frac{1}{\gamma_E} + \frac{1}{\gamma_I}$ (T7c)	$\mathcal{L}[g](\lambda) = \frac{1}{\ell\lambda+1} \cdot \frac{1}{\lambda+1}$ (T8c)
$\frac{dI}{dt} = \gamma_E E - \gamma_I I$ (T5c)	$\frac{dY_I}{d\tau} = \frac{1}{\ell} Y_E - Y_I$ (T6c)	$\ell = \gamma_I / \gamma_E$ (T7d)	$\lambda^\pm = \frac{2(\mathcal{R}_0 x^\pm - 1)}{\sqrt{(1-\ell)^2 + 4\ell\mathcal{R}_0 x^\pm} + (1+\ell)}$ (T8d)
	$\iota = \mathcal{R}_0 X Y_I$ (T6d)	$\mathcal{R}_0 = \beta \gamma_I^{-1}$ (T7e)	

Renewal equation

Dimensionless renewal equation	For general $g(\alpha)$	For Gamma $g(\alpha)$ [$a = \frac{\mu^2}{\sigma^2}, b = \frac{\mu}{\sigma^2}$]
$\frac{dX}{d\tau} = -X(\tau)F(\tau)$ (T9a)	$\mu = \int_0^\infty \alpha g(\alpha) d\alpha$ (T10a)	$g(\alpha) = \frac{b^a}{\Gamma(a)} \alpha^{a-1} e^{-b\alpha}$ (T11a)
$F(\tau) = \mathcal{R}_0 \int_{-\infty}^\tau X(\alpha) F(\alpha) g(\tau - \alpha) d\alpha$ (T9b)	$\tau = t/\mu$ (T10b)	$\mathcal{L}[g](\lambda) = \left(\frac{b}{\lambda+b}\right)^a$ (T11b)
$\iota = X F$ (T9c)	$\frac{1}{\mathcal{R}_0 x^\pm} = \mathcal{L}[g(t)](\lambda^\pm)$ (T10c)	$\lambda^\pm = b((\mathcal{R}_0 x^\pm)^{1/a} - 1)$ (T11c)

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